

SPGD-TME V2 rebuttal lighthouse

Date: 2026-09-12

Purpose: Evidence-linked control registry for the Reviewer 1 and Reviewer 2 point-to-point responses.

Contribution and provenance decision

SPGD-TME is presented as a standalone Methods contribution and an independent public repository. The scientific contribution comprises: (i) the locked malignant-coordinate KEEP/ABSTAIN reportability operator, (ii) the Patient_ID-disjoint donor construction, (iii) the explicit decision contract for withholding a non-reportable malignant coordinate, and (iv) the cross-library sensitivity, uncertainty, renormalization, and reproducibility audits added in V2.

The released SPGD `build_v4` implementation is identified only as the fixed computational baseline used for the composition fit. Its sibling-repository import is disclosed as a software packaging boundary and a provenance record. It does not define the scientific identity of SPGD-TME, and the manuscript no longer describes this work as a wrapper around an unpublished manuscript or as contingent on peer-review status elsewhere.

Common issue across both reviews

Reviewer 1's reproducibility/usability observations and Reviewer 2's comments on the baseline citation, data licensing, and citation status ask the same underlying question: which claims are new, which inputs are externally sourced, and which artifacts permit independent re-execution? V2 answers this with three explicit layers: scientific contribution, independent archive, and fixed-baseline/data provenance. The answer is now stated in the manuscript, both point-to-point letters, the public-repository metadata, and the submission capture.

The NanoString issue is a **NanoString Data License Agreement** governing redistributed CosMx data releases. It is a data-redistribution and derived-artifact-permission question, not a model-training-license claim. The Zenodo record and attachment attached-file identity are recorded. Because the agreement body could not be inspected clause by clause in the current network path, V2 makes no affirmative claim that a particular derived matrix may be redistributed; the final data statement remains conditional on the license text.

Reviewer 1 registry

Point	Disposition	Added result / evidence	V2 action
Benefits of KEEP/ABSTAIN and patient-disjoint donors	Preserved and expanded	Frozen <code>c*=0.80</code> , withheld malignant coordinate, donor-pair status, algorithm panel	Explain the two operators as the contribution and link locks, tests, and archive

Point	Disposition	Added result / evidence	V2 action
“More practice” / “more laboratory support”	Recast as usability feedback	Quick-start, environment, hardware, timing, refusal behavior, troubleshooting, and no-new-specimen statement	Add computational usability documentation; do not treat this as a wet-lab requirement
Reproducibility of the public package	Substantially resolved	Historical baseline, V2 mirror, frozen-decision records, tests, plot-data integrity records, PDF capture, integrity record	Keep v0 immutable; release V2 as a separate review build

Reviewer 2 registry

Point	Status	Added result / evidence	Remaining boundary
R2-1 baseline citation and contribution boundary	96%	Standalone-method wording; released-baseline statement; source-level constants and held-out-g audit; software-archive citation status	External baseline remains an explicit software provenance boundary
R2-2 cutoff origin and sensitivity	86%	Eight historical calls stable over 0.75–0.85; complete-scan maxima 0.847/0.791/0.854/0.905 for NSCLC/CRC/HCC/PDAC	No retrospective ROC derivation is claimed; the historical lock is retained
R2-3 neighbor-pair selection	65%	Formal eligible-set reconstruction; CRC 15-neighbor recovery; HCC preference mutation identified; historical/max-rule comparison	Author-lock one uniform rule and rerun downstream fits before final submission

Point	Status	Added result / evidence	Remaining boundary
R2-4 platform-independence claim	90%	Continuous cosine audit; CosMx designated range 0.506–0.862 with both decisions present; claim softened	Eight-library design cannot establish a causal platform effect
R2-5 uncertainty comparison	92%	Matched saved-output audit: malignant RMSE SPGD 0.154, cell2location 0.249, RCTD 0.121; native cell2location widths recorded	No fresh RCTD/cell2location refit; diagnostics remain method-specific
R2-6 ABSTAIN renormalization	98%	Simplex error 1.1×10^{-15} ; conditional neighbor bias/RMSE bootstrap audit	Truth is cell-count based; RNA-weighted truth unavailable
R2-7 truth construction	100%	Cell-count estimand explicitly labelled; transcript-mass limitation added	RNA-weighted reconstruction requires licensed source mapping
R2-8 epsilon	100%	epsilon=1e-12 recorded from source audit	None for current archive
R2-9 held-out g	100%	Clipped mean of two cross-half held-out relative Poisson-deviance reductions	None for current archive
R2-10 timing	99%	Intel Core Ultra 9 275HX, 24 logical CPUs, 62 GiB RAM, Python 3.13.7; repeated-run summary	Timing remains machine-specific
R2-11 reference-cell floor	98%	Twelve downstream fits across three edges and four floors; edge-specific RMSE/PCC sequences integrated	Scope is bounded to the saved donor construction
R2-12 numerical precision	99%	Manuscript-wide prose precision pass; no TeX decimal beyond four places	Exact values intentionally remain in machine-readable evidence

Point	Status	Added result / evidence	Remaining boundary
R2-13 bootstrap subsample	98%	400-spot design choice and bootstrap evidence documented	Full-n rerun is not required for the stated design but can be added
R2-14 neighbor fractions	82%	Fractions promoted into the V2 crosswalk and response	Main-text table/panel placement should be finalized with journal float package
R2-15 length/scope	70%	Review build contains 14 figures and 3 tables, with new evidence panels	Final journal package must decide which descriptive panels move to supplement
R2-16 data licensing	40%	NanoString Data License Agreement named; Zenodo attachment, attached-file identity, and record captured	Clause-level permission cannot be asserted until the agreement body is inspected
R2-17 citation status	95%	Reference labelled as public software archive/unpublished; no peer-review implication	Verify metadata again at final submission
R2-18 internal consistency	98%	Spot-checked values remain consistent after V2 edits	Final PDF/table cross-check remains a release gate

Added scientific interpretation carried into the manuscript

The thin-response audit triggered six substantive additions rather than a prose-only expansion:

- **Comparator distinction (R2-5).** The matched 400-spot audit now separates predictive error and interval width from the new reportability question. SPGD, cell2location, and RCTD malignant RMSEs are 0.154, 0.249, and 0.121; cell2location exposes interval widths, while the saved RCTD output lacks a matched interval/confidence field. The manuscript therefore presents the contribution as an explicit missing-coordinate decision layer compatible with composition estimators, not as a universal RMSE ranking (Discussion; Fig. 14A–B).
- **Conditional simplex (R2-6).** The forced-ABSTAIN audit shows closure to 1.1×10^{-15} but positive collinear-neighbor bias and higher conditional RMSE. Methods Eq. 9, Results, Discussion, and Fig. 14A–D now state that ABSTAIN preserves a conditional non-malignant composition and must not be scored as malignant zero.
- **Numerical contract (R2-8).** $\epsilon = 1 \times 10^{-12}$ is now explained as a fixed stability constant distinct from $c^* = 0.80$. This makes entropy weights and cutoff-adjacent calls

reproducible (Methods Eq. 1; exact-value inventory).

- **Transfer safeguard (R2-9).** The two-way held-out Poisson-deviance construction of g is explained as a self-limiting transfer fraction, so platform correction is not granted full weight from in-sample improvement alone (Methods Eq. 2).
- **Computational burden (R2-10).** Five warm runs (22.565 ± 2.861 s, CV 12.68%) and the 0.455-s/1,000-application refusal timing distinguish baseline fit cost from the inexpensive new decision layer (Methods Timing; timing table).
- **Readable yet exact reporting (R2-12).** Narrative values are rounded to three or four significant figures, while exact threshold margins, bootstrap values, and invariants remain in machine-readable evidence. This improves reader-facing clarity without weakening auditability.

These additions preserve the positive architecture of the paper: the two locked operators remain the central contribution, while the revision now explains their statistical estimand, computational cost, numerical contract, and relationship to established diagnostics.

New bounded floor evidence

The floor extension now contains twelve real downstream fits: all three directed edges at floors 25, 50, 75, and 100. Floors 50, 75, and 100 are identical within each edge because support is unchanged. Malignant RMSE changes from 0.159 to 0.163 for A-from-B, 0.102 to 0.103 for B-from-A, and 0.138 to 0.122 for D-from-C when moving from floor 25 to 50; the corresponding PCC changes are 0.853 to 0.850, 0.976 to 0.975, and 0.941 to 0.945. These results support a measured edge-dependent sensitivity statement for this donor construction; they do not assert universal floor invariance.

Low-resolution gates that must not be hidden

The items below 90% are intentionally not marked complete. They are the remaining scientific or editorial gates, not reasons to weaken the response:

1. **Neighbor selection (65%)** — author-lock one uniform rule, publish the complete per-library/per-neighbor table, and rerun any downstream fit affected by the corrected gate.
2. **Reference-cell floor (98%)** — twelve bounded fits are complete at floors 25/50/75/100 across all three directed edges; report the edge-specific sensitivity and retain the donor-construction scope.
3. **Precision (98%)** — the manuscript-wide prose pass is complete; threshold decisions use sensible significant figures while exact values remain in tables and machine-readable artifacts.
4. **Neighbor-fraction visibility (82%)** — finalize the main-text panel/table placement and ensure its caption is standalone.
5. **Scope/float package (70%)** — keep the review build expansive for inspection, then prepare a journal-targeted figure/supplement partition without deleting the expanded evidence.
6. **License (40%)** — inspect the attached NanoString agreement clause by clause when the PDF is accessible; until then retain conditional data-availability wording.
7. **Uncertainty comparison (92%)** — the matched saved-output audit is complete for the available inputs; retain the method-specific diagnostic boundary and do not equate unlike uncertainty fields.
8. **Cutoff provenance (86%)** — retain the pre-sweep lock and sensitivity result, while explicitly stating that no empirical ROC selection was performed.

Release rule

A point is marked resolved only when the evidence record, manuscript text, point-to-point response, portal-clean text, PDF, and integrity record agree. The historical V0/V1 artifacts remain untouched. The V2 review build is therefore reviewable now, while the seven gates above remain visible for the next analysis pass.

New gate-input consistency audit (C03)

The released reportability operator consumes the normalized mean reference profiles (P); the platform-corrected working signature (S) is reserved for the Poisson composition fit. This distinction is now stated explicitly in Methods and tested as a regression contract. A bounded deterministic reconstruction on 400 spots per available benchmark compared (P) and (S): `realgt` and `realgt3` retained the same binary calls, while `openST` was a boundary discrepancy (P=0.972678, ABSTAIN; reconstructed S=0.693686, KEEP under the bounded audit settings). The historical primary calls are unchanged. We retain this discrepancy as a transparent implementation-scope finding and do not substitute (S) into the locked gate without a new author lock and dependent refit.

This result strengthens the reproducibility response because it identifies the precise semantic boundary between reportability and composition fitting. It also prevents the manuscript from implying that the working signature, rather than the implemented mean-profile input, defined the historical gate.